

DIVYANSH GUPTA

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SUMMARY

Software engineer with strong CS fundamentals (Data Structures & Algorithms in C++) who ships production systems end to end — from a low-latency C++ matching engine to full-stack platforms (FastAPI + Next.js) and LLM-integrated multi-agent AI — with security and correctness engineered in from the first commit. Deepest in backend and applied AI, fluent across the full stack. Three internships, a patent pending, a national hackathon finalist, and four live deployed products.

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, JavaScript, SQL

Backend & Systems: FastAPI, REST API Design, Microservices, Distributed Systems, System Design, Celery + Redis (task queues), Concurrency, Performance Optimization

AI / ML: LLM Integration (Claude, Groq, Llama 3.3), CrewAI Multi-Agent Systems, spaCy, sentence-transformers, scikit-learn

Databases: PostgreSQL, MySQL, SQLAlchemy (async), Elasticsearch, Redis

DevOps & Cloud: Docker, Kubernetes, CI/CD (GitHub Actions), Linux, Vercel, pytest (unit + integration testing)

Frontend & Security: React, Next.js, TailwindCSS, Three.js/WebGL, OWASP Top 10, Threat Modeling, Smart-Contract Auditing

EXPERIENCE

Backend Developer

Jun 2026

XTRAGRAD-AI – AI-HR Recruitment Simulator

- Architected a 50-endpoint FastAPI backend (Admin + Candidate dashboards) with async SQLAlchemy, PostgreSQL, and JWT auth, powering an AI-driven recruitment platform end to end.
- Engineered a distributed Celery + Redis task queue that moved long-running AI inference out of the request path — keeping APIs instantly responsive while scaling background processing under load.
- Integrated an LLM inference microservice via a contract-driven HTTP client with local-model fallback; owned a 50+ test pytest suite (unit, integration, and end-to-end) gating every deploy in CI.

Blockchain Developer

May 2025 – Jun 2025

IBM

- Audited Ethereum smart contracts for reentrancy, access-control, and gas-misuse defects, running iterative adversarial testing to resolve high-stakes correctness issues before immutable deployment.

Frontend Developer

Jan 2024 – Mar 2024

1Stop.ai

- Built and optimized responsive UIs within an Agile team, translating business requirements into effective, user-facing product solutions with strict input validation and secure coding practices.

PROJECTS

Order-Book Matching Engine | C++, STL, Data Structures | [GitHub](#)

2026

- Built the core of an exchange in modern C++ — a limit order book with price-time priority, IOC/FOK order types, self-trade prevention, and O(1) cancel — matching ~3.5M orders/sec at ~280 ns/order, backed by a 35-check test suite and GitHub Actions CI.

CivicShield | Next.js, FastAPI, Python, Docker, Selenium | [GitHub](#) | [Live](#) | *Hack Energy 2.0 Finalist*

2026

- Built a full-stack web-security platform (Next.js + FastAPI): SQL-injection/XSS fuzzing, access-control auditing, and an ML phishing classifier, backed by a Dockerized CI test pyramid (pytest, Postman/Newman, Selenium E2E) gating every push.

TrustSphere AI | Python, FastAPI, HuggingFace, Next.js | [GitHub](#) | *In Development*

2026

- Building a multi-agent deepfake and forgery detector: four forensic agents (image, audio, metadata, security) over HuggingFace ViT/Wav2Vec2, fused by a weighted engine into one authenticity verdict, with blockchain-anchored evidence reports.

ChainAudit | Next.js, TypeScript, FastAPI, Python, Docker | [GitHub](#) | [Live](#)

2026

- Designed and shipped a live full-stack platform automating Solidity smart-contract audits — a Next.js interface over a FastAPI + Slither backend — with severity-weighted risk scoring, deduplicated findings, and PDF reporting, deployed via Dockerized CI/CD.

PATENT & ACHIEVEMENTS

Patent (pending): “Automated Processing and Security Risk Analysis of Enterprise Information Assets Using Zero-Shot Agentic AI with Autonomous Cybersecurity Response”

NPTEL Gold Medal — scored 100/100 in Wildlife & Ecology | **State-level Table Tennis Champion**

EDUCATION & LEADERSHIP

Vellore Institute of Technology

Expected 2027

Bachelor of Computer Science and Engineering – CGPA: 8.87 / 10

- Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Cryptography.

General Operations & Management Head – VIT Blockchain Community Club

2024 – Present

- Lead operations for the college blockchain community; mentor 150+ students in software, security, and decentralized systems.

CERTIFICATIONS

Google Cybersecurity Professional Certificate – Coursera | Blockchain – IBM | Python – Udemy | Web Development – Udemy